Med Aziz Hraghi

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Seeking a 6-month AI / Machine Learning PFE internship | Available Jan–Feb 2027

Profile

Final-year engineering student in Digitalisation & Data Analysis with hands-on experience building **retrieval-augmented AI applications, LangGraph workflows, and computer-vision systems**. Combines Python/FastAPI development with document retrieval, anomaly detection, and research-oriented decision support.

Experience

AI Engineer Intern — *Talan Tunisie (Summer Camp 2026)* Jul–Aug 2026 | On-site
Khatatli UrbanTwin checks parcel zoning and turns site analysis into CAD-ready plans. / **Live demo**

- Engineered a **RAG pipeline** over **8 French legal codes** and local PLU zoning rules using Mistral embeddings and Qdrant, producing regulation answers with **verifiable legal citations**.
- Built a **LangGraph multi-agent pipeline** aggregating **20+ public data sources** across climate, elevation, soil, flood risk, mobility, cadastre, and POIs to automate terrain analysis and feasibility assessment.
- Developed generative planning features combining **2D site-plan generation, DXF/GeoJSON/PDF export, and 3D digital-twin visualization**, plus a 15-minute-city accessibility evaluator spanning **9 service categories**.
- Implemented EU AI Act transparency features and authored **733 automated offline tests** within a **12-member engineering team** working on a FastAPI and React / TanStack Start stack.

Applied AI Intern — *INAT – LSTE (National Agronomic Institute of Tunisia)* Jul–Aug 2026 | Remote, part-time

- Developed a **scientific-intelligence pipeline** ingesting **ArXiv, PubMed, and Scopus**, using embedding-based deduplication, semantic retrieval, and automated summarization to support research monitoring.
- Built an **AI-assisted irrigation decision-support module** combining field-sensor ingestion, data-quality validation, and **water-balance scenario generation** for agricultural parcels.
- Orchestrated **6 specialized AI modules** across scientific watch, bibliography, project quality, decision support, and workflow coordination, using FastAPI, PostgreSQL/pgvector, and automated offline testing.

Education

National School of Advanced Sciences and Technologies of Borj Cedria (ENSTAB)
Engineering Degree in Advanced Technologies — Specialization: Digitalisation & Data Analysis (DAD) Sep 2024 – Present

Key courses: Generative AI, Advanced AI, Machine Learning, Image Processing, DevOps/MLOps, Big Data, Full-Stack Web Development, Databases, IoT, Cloud & Virtualization

Faculty of Sciences of Tunis (FST)
Preparatory Cycle: Mathematics and Physics National entrance exam for engineering schools 2024

Projects

AgenticHire — AI-Powered Recruitment Platform — *PFA* • **3rd Place · NerdData ENSIT Hackathon**
Helps candidates find jobs that match their CVs and recruiters screen applicants against role requirements.

- Architected AI-assisted recruiting workflows for candidates and recruiters, using **Mistral AI and Hugging Face models** with specialized agent routing.
- Built a multi-source job-ingestion pipeline aggregating listings from **11 platforms** through REST APIs, BeautifulSoup, and python-jobspy.
- Developed an end-to-end candidate intelligence workflow combining intent detection, CV analysis, semantic job matching, CV optimization, cover-letter generation, and recruiter-side candidate evaluation.
- Developed the application stack with a **FastAPI backend**, JWT authentication, SQLite/Pydantic data models, and a React/Vite conversational frontend.

Syntern (Stagi) — AI Internship Simulator • **2nd Place · TBS Atlas Hackathon**
Gives students a simulated internship where they practice handling tasks and feedback from different workplace roles.

- Contributed to an internship simulator featuring **4 role-specific AI agents** (Manager, Tech Lead, Client, Senior Intern) and voice interaction via ElevenLabs and Web Speech API.
- Engineered an **n8n orchestration layer** for session management, agent routing, inactivity escalation, and behavioral evaluation, scoring users across **5 professional dimensions**.
- Integrated the **n8n workflows** with a React/Vite workspace supporting communication, AI-generated Kanban tasks, Supabase persistence, and GitHub code-review automation.

GROOV — Multi-Agent AI Study Group

Turns course PDFs into interactive study sessions where AI tutors cite the material and help students spot gaps in their understanding.

- Built a local multi-agent learning platform with **6 pedagogical AI personas** powered by Ollama (LLaMA 3.2), with **4 adaptive study modes** and intelligent speaker orchestration.
- Engineered a **RAG pipeline** using FAISS and Sentence-Transformers for semantic PDF retrieval with page-level citations and document-grounded agent discussions.
- Developed multimodal learner-state adaptation by combining webcam-based emotion recognition with LLM sentiment analysis across **8 emotional states**, dynamically modifying teaching behavior.
- Implemented a Feynman-style evaluation workflow producing **0–100 feedback scores**, missing-concept detection, analogy assessment, and targeted improvement recommendations.

Vendex — AI Forensic Claims Intelligence • **Top 6 of 37 Teams · GDGC FST Hackathon**

Flags suspicious insurance claim photos and shows reviewers which image signals triggered the flag.

- Designed a **five-stage computer-vision forensic pipeline** for detecting manipulated, recycled, and AI-generated insurance-claim imagery.
- Combined **Error Level Analysis, FFT, DINOv2 embeddings, and Mahalanobis distance anomaly detection** to identify visual inconsistencies across complementary forensic signals.
- Integrated **MobileSAM segmentation** for localized damage analysis and EXIF metadata forensics, served through asynchronous FastAPI inference.
- Developed a three-agent Mistral-based reasoning layer that synthesizes forensic evidence into an **explainable fraud assessment** and automated PDF report.

SurgiGuard AI — Postoperative Recovery Monitoring • **2nd Place · HackCure 2026**

A team-built prototype for monitoring recovery at home after surgery and surfacing changes for care-team review.

- Integrated an **ESP32 wearable** that streams heart rate, blood oxygen, motion, and temperature over **MQTT**.
- Connected wearable telemetry to a **LangGraph workflow** that reviews changes in vitals and uses **rules-based escalation** to flag readings for care-team review.
- Combined a **retrieval-augmented assistant** for drafting recovery plans from medical documents with computer vision for wound photos and rehab exercises.

Skills

- **Programming:** Python, C++, Java, SQL, JavaScript, TypeScript
- **Machine Learning / Deep Learning:** PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, DINOv2, MobileSAM, Anomaly Detection, Computer Vision, NLP
- **Generative AI / Agentic AI:** LLMs, Multi-Agent Systems, LangGraph, RAG, Mistral AI, Ollama, FAISS, Qdrant, Sentence-Transformers, n8n
- **Backend / Engineering:** FastAPI, REST APIs, Pydantic, PostgreSQL/pgvector, Supabase, Docker, Git
- **Data / Scientific Computing:** NumPy, SciPy, MATLAB, Signal Processing, Geospatial Data Processing
- **Frontend / Visualization:** React, Vite, TanStack Start, Three.js, react-three-fiber, Leaflet
- **Languages:** Arabic (Native), French (Fluent), English (Fluent)

Leadership & Activities

- **ACM ENSTAB Club** Chairman — led club operations, technical training, competitive programming activities, and events | Sep 2025 – Sep 2026
- **ETC Junior Entreprise —** Project Management — coordinated project planning, team execution, client communication, and delivery tracking | Sep 2024 – Present

Certifications

Fundamentals of Deep Learning — NVIDIA — Jan 2025

AI Engineer for Data Scientists — DataCamp — In progress (expected Nov 2026)